

Every rank-one operator has an algebra preimage

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton`

theorem

Turns one represented rank-one projection into every rank-one operator using dense orbits and a closed range.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ represent the unique unitary-equivalence class of nonzero irreducible $*$ -representations. For every $x, y \in H$ there is $a \in A$ such that

$$\pi(a) = \theta_{x,y}, \quad \theta_{x,y}(z) = \langle y, z \rangle x.$$

Assumptions

The vectors x, y are arbitrary, including zero. They need not be unit vectors. The separability assumption concerns H . No unit is required in A .

Conclusion

Each rank-one operator belongs to the image of the original algebra, not merely to the image of its unitization.

Proof route

Find one rank-one projection in the range. Use irreducibility to obtain a dense original-algebra orbit, multiply around the projection, and pass to limits in both vectors.

▼ Proof steps

1. Apply A rank-one projection in a singleton model to the nonzero ordered A , separable H , and singleton π . It supplies $p \in A$ and a unit vector $e \in H$ with $\pi(p) = \theta_{e,e}$. In particular $\pi(p)e = e$. The theorem Faithfulness of a non-unital singleton model also gives injectivity of π .
2. The dense-orbit step in One rank-one projection generates all rank-one operators in the range uses irreducibility of the unitized representation. Its orbit of $e \neq 0$ is dense; every unitization element has the form (λ, a) and acts by

$$\widehat{\pi}(\lambda, a)e = \lambda e + \pi(a)e = \pi(\lambda p + a)e.$$

Thus already $\{\pi(a)e : a \in A\}$ is dense in H . This calculation is why the preimages below stay in A .

3. For $a, b \in A$ and $z \in H$, multiplicativity and adjoints give

$$\pi(apb^*)z = \pi(a)\theta_{e,e}\pi(b)^*z = \langle e, \pi(b)^*z \rangle \pi(a)e = \langle \pi(b)e, z \rangle \pi(a)e.$$

$$\text{Hence } \pi(apb^*) = \theta_{\pi(a)e, \pi(b)e}.$$

4. Because π is injective, it is isometric and has norm-closed range. Approximate arbitrary x, y by vectors $\pi(a)e, \pi(b)e$ in the dense orbit. The estimates $\|\theta_{u,v}\| \leq \|u\|\|v\|$ and

$$\|\theta_{u,v} - \theta_{x,y}\| \leq \|u - x\|\|v\| + \|x\|\|v - y\|$$

show convergence of the corresponding rank-one operators. Closedness then puts $\theta_{x,y}$ in $\pi(A)$, which is the asserted existence of a .

Main citations

- Exact theorem and proof
- Faithfulness of a non-unital singleton model
- One rank-one projection generates all rank-one operators in the range
- A rank-one projection in the original algebra

Lean source signature (exact)

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theorem exists_preimage_rankOne_of_singleton [Nontrivial A]
  [PartialOrder A] [StarOrderedRing A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  ∀ x y : H, ∃ a : A, pi a = InnerProductSpace.rankOne C x y
```

In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[PartialOrder A] [StarOrderedRing A]	These typeclasses provide the usual order on a C^* -algebra, equivalently $a \leq b$ when $b - a$ is positive. They are used by the state and positivity arguments.
[TopologicalSpace.SeparableSpace H]	H is a separable complex Hilbert space.
pi : NonUnitalCStarRepresentation A H	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; its interface does not require a unit equation.
hsingle : IsSingletonIrreducibleModel.{u, v, u} pi	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
∀ x y : H, ∃ a : A	After any two vectors x, y are chosen, an element a may be chosen depending on both.

In the source	Mathematical meaning
<code>InnerProductSpace.rankOne $\mathbb{C}$ x y</code>	The entire operator $z \mapsto \langle y, z \rangle x$, with this order of x, y .
<code>pi a = InnerProductSpace.rankOne $\mathbb{C}$ x y</code>	The chosen a represents that operator on every vector $z \in H$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton`

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